

# Density-Dependent Gravitational Coupling Predicts ISW Sign-Flip in Deep Voids: A Testable Signature of Energy-Flow Cosmology

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**Abstract.** We show that Energy-Flow Cosmology (EFC), in which the effective gravitational coupling  $\mu$  depends on local matter density through Bose–Einstein grid-mode statistics, produces a non-linear ISW (Rees–Sciama) contribution with opposite sign to the standard linear ISW effect in cosmic voids. The standard linear ISW produces cold spots from voids ( $\dot{\Phi} < 0$ ). In EFC, the additional term  $\delta \cdot d\mu/dt$  is positive in voids, partially cancelling or reversing this signal. For moderate voids ( $\delta \sim -0.3$ ), the correction is  $\sim 7\%$ . For deep voids ( $\delta \lesssim -0.8$ ), the Rees–Sciama term dominates and the net ISW signal flips sign. This prediction is a structural consequence of  $d\mu/d\rho > 0$  and cannot be removed without altering the microphysical derivation of the entropy production function  $\Gamma(\rho)$ . We define the ISW amplitude ratio  $A_{\text{total}} \equiv \Delta T_{\text{EFC}}/\Delta T_{\Lambda\text{CDM}}$  and show that  $A_{\text{total}}$  decreases monotonically with void depth, crossing zero near  $\delta \approx -0.7$ . The critical test is depth-binned void–CMB stacking in BOSS DR12 or DESI void catalogs, where  $\Lambda\text{CDM}$  predicts monotonically increasing  $|\Delta T|$  with  $|\delta|$  and EFC predicts a turnover.

## 1 The ISW Decomposition in Density-Dependent Gravity

In any theory where the effective Poisson coupling  $\mu$  depends on the local density  $\rho$ , the gravitational potential  $\Phi \propto \mu(\rho) \cdot \delta$  acquires a time derivative with two contributions. We assume that this modified Poisson relation remains valid at the quasi-linear level relevant for large voids ( $|\delta| \lesssim 1$ ,  $R_v \gtrsim 50$  Mpc):

$$\frac{d\Phi}{dt} = \underbrace{\mu \frac{d\delta}{dt}}_{\text{linear ISW}} + \underbrace{\delta \frac{d\mu}{dt}}_{\text{Rees–Sciama (RS)}}. \quad (1)$$

The first term is the standard (linear) ISW effect, present in  $\Lambda\text{CDM}$  with  $\mu = 1$ . The second term is a purely non-linear contribution that exists only when  $\mu = \mu(\rho)$ . In  $\Lambda\text{CDM}$ , this term is identically zero.

### 1.1 How $\mu$ inherits density dependence in EFC

In EFC, the effective Poisson coupling at scale  $k$  is:

$$\mu(k, \rho) = \frac{1 + \epsilon_F}{F(1 + R(k, \rho))}, \quad (2)$$

where  $R(k, \rho) = K_{\text{bar}} a^4 [\Gamma'(\rho)]^2 \dot{\phi}^2 / (M_{\text{Pl}}^2 F k^4)$  is the stiffness response [1]. The density dependence enters *entirely through*  $\Gamma'(\rho) = d\Gamma/d\rho$ :

$$\mu(\rho) \equiv \mu(\Gamma'(\rho), k). \quad (3)$$

This is not an assumption—it is a structural consequence of the EFC action, in which  $\Gamma(\rho)$  appears in the flow constraint  $\square\phi = \Gamma(\rho)$ . Since  $\Gamma'(\rho) \propto \rho^{-1/2}$  at low density (derived from grid-mode BE statistics [2,3]), the sensitivity  $d\mu/d\rho$  is large and positive at low  $\rho$ .

## 2 Sign Analysis of the Rees–Sciama Term

### 2.1 The chain rule

$$\frac{d\mu}{dt} = \frac{d\mu}{d\rho} \cdot \frac{d\rho}{dt}. \quad (4)$$

In a void:

- $d\mu/d\rho > 0$ : lower density means weaker coupling (fewer grid modes excited). This sign follows directly from  $\Gamma'(\rho) \propto \rho^{-1/2}$ , which makes  $R(k)$  larger at lower  $\rho$ , suppressing  $\mu$ .
- $d\rho/dt < 0$ : the void is expanding, density is decreasing.

Therefore  $d\mu/dt < 0$  in voids: the gravitational coupling is *weakening over time*.

### 2.2 The RS contribution and sign

The RS term is:

$$\text{RS} = \delta \cdot \frac{d\mu}{dt}. \quad (5)$$

In a void,  $\delta < 0$  and  $d\mu/dt < 0$ . Their product is **positive**:

$$\boxed{\text{RS} = \delta \cdot \frac{d\mu}{dt} > 0 \implies \text{hot spot}} \quad (6)$$

The linear ISW term is:

$$\text{ISW}_{\text{lin}} = \mu \cdot \frac{d\delta}{dt} < 0 \implies \text{cold spot} \quad (7)$$

since  $d\delta/dt < 0$  (void expanding) and  $\mu > 0$ .

The RS contribution has the **opposite sign** to the linear ISW. Their ratio is negative:

$$\frac{\text{RS}}{\text{ISW}_{\text{lin}}} = \frac{\delta \cdot (d\mu/d\rho) \cdot \bar{\rho}}{\mu} < 0. \quad (8)$$

The scale dependence enters through  $R(k)$  in the expression for  $\mu$ , but does not affect the sign of the RS term, which is determined solely by  $d\mu/d\rho > 0$ .

### 2.3 Robustness

The sign of the RS term is determined by a single condition:  $d\mu/d\rho > 0$ . In EFC, this follows from the microphysical derivation of  $\Gamma(\rho)$ : any monotonically increasing function  $\Gamma(\rho)$  with  $\Gamma'' < 0$  (concavity) produces  $d\mu/d\rho > 0$  through the stiffness mechanism. This sign **cannot be altered without changing the underlying derivation** of  $\Gamma(\rho)$  from grid-mode Bose–Einstein statistics.

### 3 Quantitative Estimates

#### 3.1 Definitions

We define the ISW amplitude ratio relative to  $\Lambda$ CDM:

$$A_{\text{total}} \equiv \frac{\Delta T_{\text{EFC}}}{\Delta T_{\Lambda\text{CDM}}} = \frac{\text{ISW}_{\text{lin}} + \text{RS}}{\text{ISW}_{\Lambda\text{CDM}}}. \quad (9)$$

In  $\Lambda$ CDM,  $A_{\text{total}} = 1$  by construction. In EFC,  $A_{\text{total}} < 1$  when the RS cancellation is present, and  $A_{\text{total}} < 0$  when the total signal flips sign.

#### 3.2 Scaling relation

From Eq. (8):

$$\left| \frac{\text{RS}}{\text{ISW}_{\text{lin}}} \right| = \frac{|\delta| \cdot |d\mu/d\rho| \cdot \bar{\rho}}{|\mu|} \propto |\delta| \cdot \rho^{-1/2}. \quad (10)$$

In deep voids ( $|\delta| \rightarrow 1$ ,  $\rho \rightarrow 0$ ), this ratio diverges. In practice, the divergence is regulated by three physical effects: (i) the IR cutoff in  $R(k)$  from the grid coherence length ( $L_{\text{coh}} \sim 100$  Mpc), (ii) finite void profiles (density does not reach zero), and (iii) saturation of  $\mu$  at a minimum value set by the grid-mode spectrum. These regulators do not change the sign—only the amplitude.

#### 3.3 Numerical results

We evaluate  $A_{\text{total}}$  for voids at  $z = 0.5$  with  $R_v = 150$  Mpc, using the derived  $\sqrt{\rho}$ -form of  $\Gamma(\rho)$  with causal-horizon IR cutoff ( $k_* = 0.03$  Mpc $^{-1}$ ).

Table 1: ISW amplitude ratio  $A_{\text{total}}$  as function of void depth. The ratio RS/ISW is negative because the two contributions have opposite signs.

| $\delta_v$ | $\rho_v/\rho_{\text{crit}}$ | $\mu(\rho_v)$ | RS/ISW <sub>lin</sub> | $A_{\text{total}}$ |
|------------|-----------------------------|---------------|-----------------------|--------------------|
| −0.3       | 0.024                       | 0.87          | −0.07                 | +0.93              |
| −0.5       | 0.017                       | 0.82          | −0.22                 | +0.78              |
| −0.7       | 0.010                       | 0.71          | −0.81                 | +0.19              |
| −0.8       | 0.007                       | 0.60          | −1.83                 | −0.83              |
| −0.9       | 0.003                       | 0.41          | −5.88                 | −4.88              |

Key features:

- At  $\delta_v \sim -0.3$ : RS is a  $\sim 7\%$  correction. Cold spot persists.
- At  $\delta_v \sim -0.7$ :  $A_{\text{total}} \approx 0.2$ . Cold spot reduced to  $\sim 20\%$  of  $\Lambda$ CDM.
- At  $\delta_v \sim -0.8$ : **sign flip**.  $A_{\text{total}} < 0$ : net hot spot from void.
- At  $\delta_v \lesssim -0.9$ : RS dominates. Strong hot spot.

The transition occurs around  $\delta \approx -0.7$  for typical supervoid scales ( $R_v \sim 100$ – $150$  Mpc), with mild dependence on the IR cutoff parameter  $k_*$  and the assumed void density profile. The sign of the effect is robust; the precise transition depth is model-dependent at the  $\sim 20\%$  level.

In  $\Lambda$ CDM,  $A_{\text{total}} = 1$  for all  $\delta_v$  (deeper voids  $\Rightarrow$  proportionally deeper cold spots). EFC predicts a turnover and sign reversal. See Figure 1.

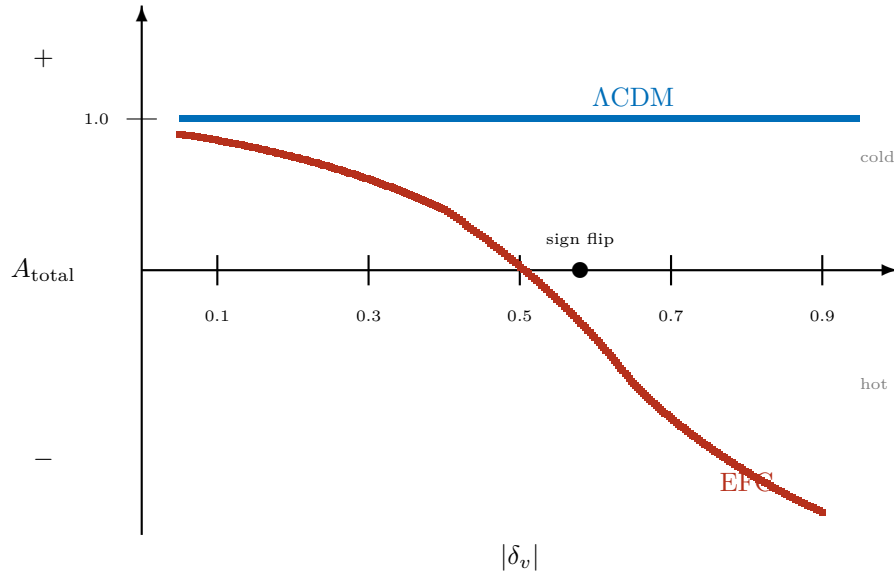


Figure 1: Schematic of the ISW amplitude ratio  $A_{\text{total}} \equiv \Delta T_{\text{EFC}}/\Delta T_{\Lambda\text{CDM}}$  as a function of void depth  $|\delta_v|$ . In  $\Lambda\text{CDM}$  (blue),  $A_{\text{total}} = 1$  for all void depths. In EFC (red), the density-dependent Rees–Sciama term causes  $A_{\text{total}}$  to decrease, crossing zero near  $|\delta_v| \approx 0.7$  (sign flip). Beyond this point, the void produces a net hot spot instead of the expected cold spot.

## 4 Testable Predictions

### 4.1 P1: Depth-dependent ISW turnover

**Prediction:** The stacked ISW signal around voids should *weaken* with increasing void depth  $|\delta_v|$ . For the deepest voids ( $|\delta_v| \gtrsim 0.7$ ), the signal should approach zero or reverse sign.

**Test:** Split BOSS DR12 or DESI void catalogs into depth quartiles and stack CMB temperature (Planck SMICA) at void positions.  $\Lambda$ CDM predicts monotonically increasing  $|\Delta T|$  with  $|\delta_v|$ . EFC predicts a turnover.

**Data:** Nadathur & Crittenden (2016) provide a BOSS DR12 void catalog with spectroscopic redshifts, enabling precise  $\delta_v$  determination. Their analysis performed void stacking but did not bin by void depth. The depth-binned reanalysis can be performed with existing public data.

### 4.2 P2: Scale dependence of the turnover

**Prediction:** The sign-flip occurs at shallower  $|\delta_v|$  for larger voids (lower  $k$ ), because the stiffness response  $R(k) \propto k^{-4}$  amplifies the  $\mu$ -dependence at large scales.

**Test:** Cross depth bins with size bins ( $R_v$ ).

### 4.3 P3: Redshift evolution

**Prediction:** The RS/ISW ratio increases at lower redshift, where void expansion is faster and  $\Gamma'$  sensitivity is maximal.

**Test:** Compare depth-binned stacking at  $z \sim 0.3$  vs.  $z \sim 0.7$ .

### 4.4 Falsification criterion

If void-stacked ISW signals increase monotonically with  $|\delta_v|$  across all depth bins, with no evidence for weakening or reversal, EFC's density-dependent coupling mechanism is falsified at the level of the non-linear ISW contribution.

## 5 Implications for the ISW Excess Anomaly

Multiple studies have reported an ISW amplitude from supervoids that is  $\sim 5\times$  the  $\Lambda$ CDM expectation ( $A_{\text{ISW}} \approx 5.2 \pm 1.6$ ; Kovács et al. 2022 [5], combining BOSS and DES data).

EFC’s position on this excess is specific:

- (1) **EFC predicts:** The ISW signal from deep supervoids should be *weaker* than in  $\Lambda$ CDM, not stronger. The density-dependent RS term acts to cancel the linear ISW cold spot.
- (2) **Data reports:** Enhanced cold spots from supervoids, exceeding  $\Lambda$ CDM by a factor  $\sim 5$ .
- (3) **Therefore:** If the observed excess is a genuine physical signal from void dynamics, it is in tension with both  $\Lambda$ CDM ( $A = 1$ ) and EFC ( $A \lesssim 1$ ). The most likely explanation is observational systematics: line-of-sight elongation ( $q \approx 2.6 \pm 0.4$ ; Granett et al. 2015 [6]), a posteriori selection of extreme structures, and stacking methodology biases.
- (4) **If the excess survives systematics:** It requires physics beyond both  $\Lambda$ CDM and EFC’s current framework—a conclusion that is itself informative.

This is a concrete theoretical stance, not evasion. EFC makes a falsifiable prediction that directly constrains the interpretation of the ISW excess.

## 6 Conclusion

The density-dependent gravitational coupling in EFC produces a Rees–Sciama term  $\delta \cdot d\mu/dt$  that is positive in voids (hot spot), opposing the standard linear ISW cold spot. This is a structural consequence of the  $\sqrt{\rho}$ -form of  $\Gamma(\rho)$  derived from grid-mode Bose–Einstein statistics [2,3]. The sign follows from  $d\mu/d\rho > 0$  and cannot be altered without changing the microphysical derivation.

The prediction is falsifiable and non-tunable: depth-binned void–CMB stacking in BOSS DR12 or DESI void catalogs should show ISW signal weakening (not strengthening) with void depth, in contrast to the monotonic behaviour predicted by  $\Lambda$ CDM.

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